

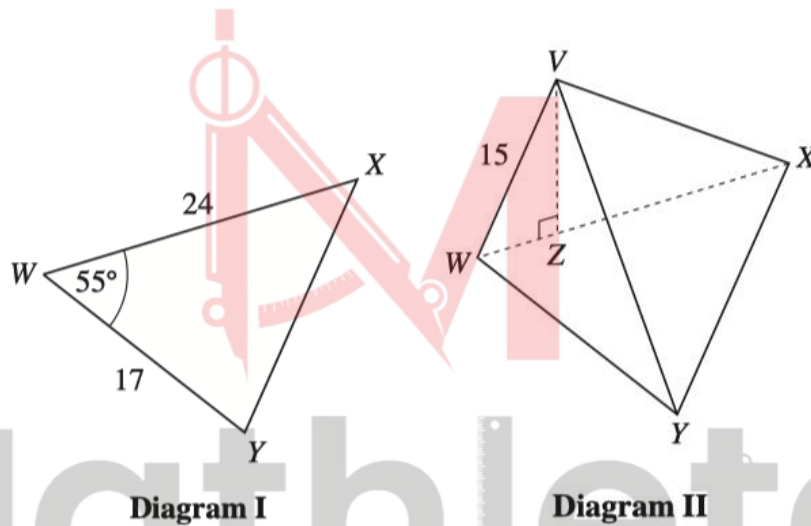
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Worksheet 2b

Pyramid

1. S10/qp21/q12(b)(ii)

[Volume of a pyramid = $\frac{1}{3} \times \text{base area} \times \text{height}$]



The triangle WXY shown in Diagram I forms the horizontal base of the triangular pyramid VWXY, shown in Diagram II. The vertex V is vertically above Z, a point on WX. $WV = 15\text{ cm}$ and $WZ = \frac{1}{4}WX$.

- Calculate VZ. [2]
- Hence find the volume of the pyramid. [1]

2. W10/qp22/q9(a)(b)

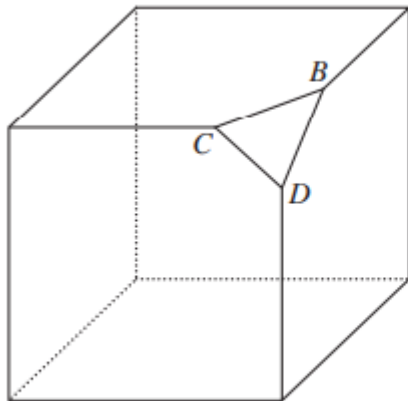


Diagram I

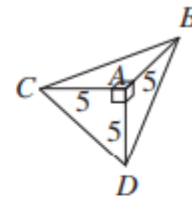


Diagram II

Diagram I shows a cube with a triangular pyramid removed from one vertex. This triangular pyramid ABCD is shown in Diagram II. $AB = AC = AD = 5$ cm.

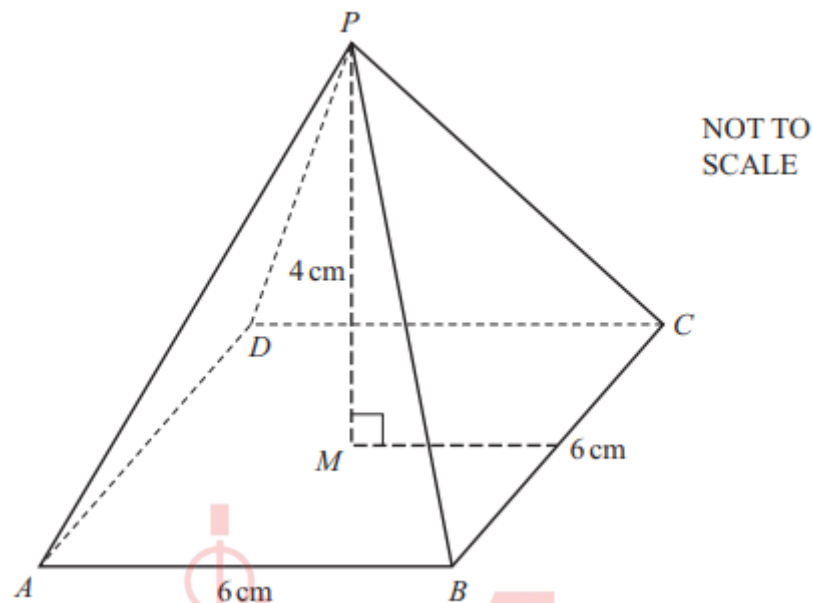
- State the height of this pyramid when the base is triangle ABD. [1]
- [The volume of a pyramid = $\frac{1}{3} \times \text{area of base} \times \text{height}$]

Calculate

- the volume of the pyramid, [2]
- the area of triangle BCD, [3]
- the height of the pyramid when the base is triangle BCD. [3]

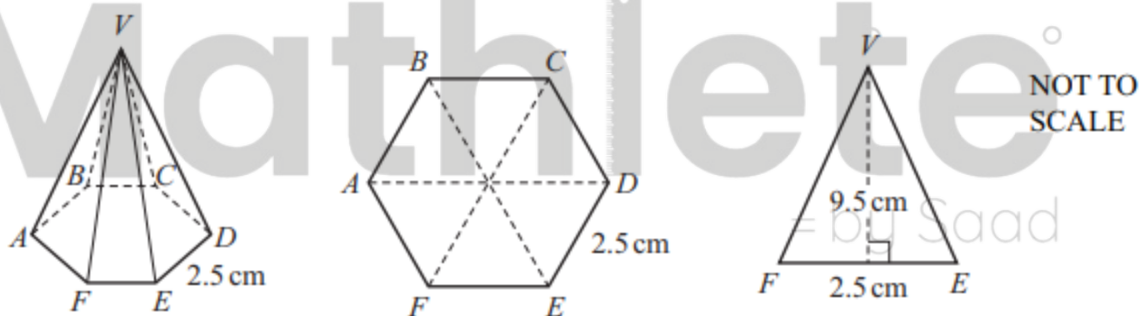
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3. S11/qp23/q21 0580



The diagram shows a pyramid with a square base $ABCD$ of side 6 cm . The height of the pyramid, PM , is 4 cm , where M is the centre of the base. Calculate the total surface area of the pyramid. [5]

4. S11/qp42/q7(a) 0580



A solid pyramid has a regular hexagon of side 2.5 cm as its base. Each sloping face is an isosceles triangle with base 2.5 cm and height 9.5 cm . Calculate the total surface area of the pyramid. [4]

5. W11/qp22/q7(b)

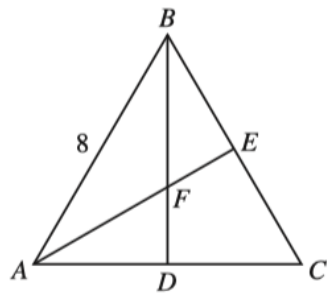


Diagram I

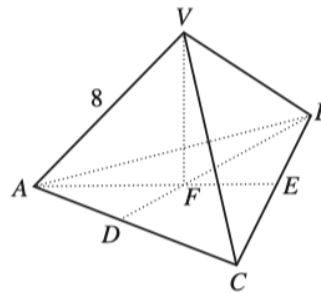
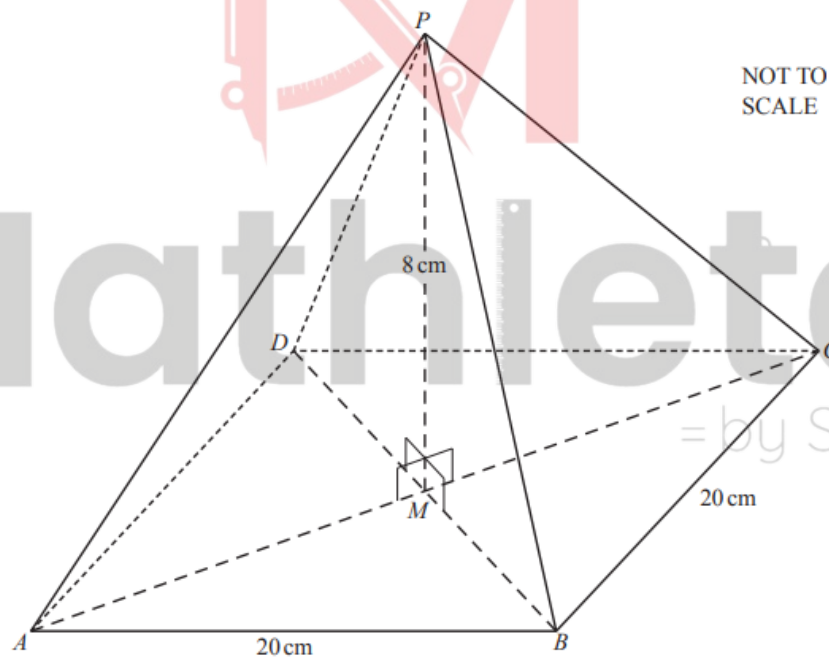


Diagram II

The equilateral triangle of side 8 cm in Diagram I forms the base of the triangular pyramid VABC in Diagram II. The vertex V is vertically above F. $VA = VB = VC = 8$ cm.

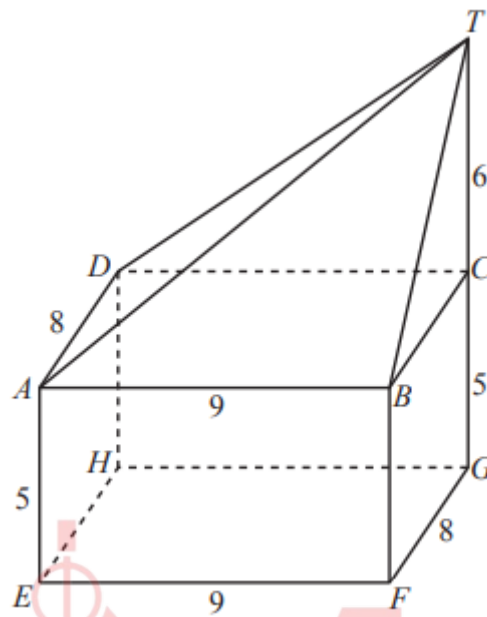
- Calculate the surface area of the pyramid. [1]
- Calculate the volume of the pyramid. [3]

6. S15/qp23/q18 (0580)



The diagram shows a solid pyramid on a square horizontal base ABCD. The diagonals AC and BD intersect at M. P is vertically above M. $AB = 20$ cm and $PM = 8$ cm. Calculate the total surface area of the pyramid. [5]

7. W16/qp21/q6



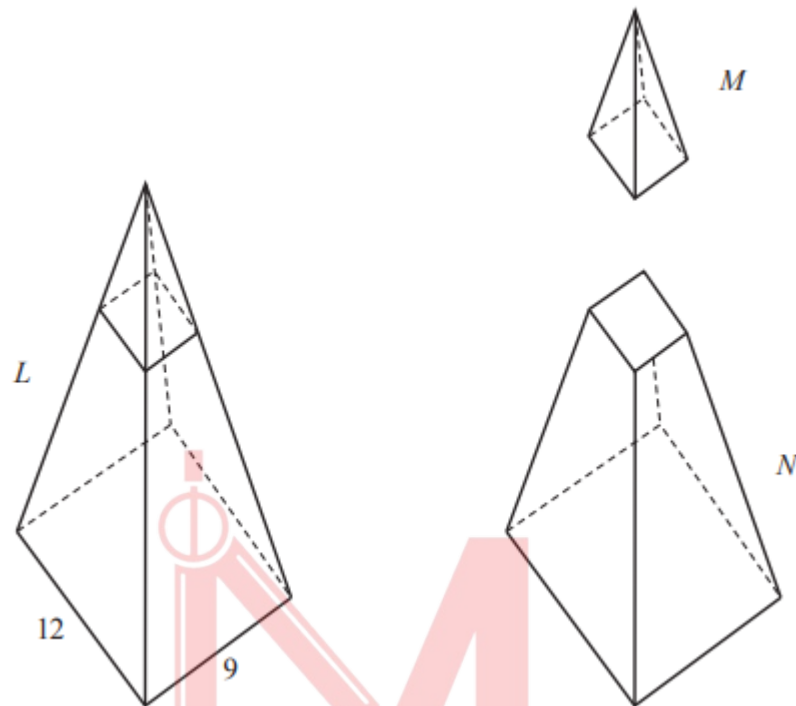
The four walls of a building are faces of a cuboid ABCDEFGH. T is vertically above C and G, so $\widehat{ABT} = \widehat{ADT} = 90^\circ$. The cuboid has length 9m, width 8m and height 5m. TC = 6m.

- Calculate the length of DT. [2]
- The roof is formed by four triangles, ABT, BCT, CDT and DAT. Calculate the total surface area of the roof. [3]
- [The volume, V , of a pyramid is $V = \frac{1}{3} \times \text{area of base} \times \text{height}$] Calculate the total volume of the building. [2]
- Calculate the angle of elevation of T from H. [3]

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8. S17/qp11/q15

[Volume of a pyramid = $\frac{1}{3} \times \text{base area} \times \text{perpendicular height}$]

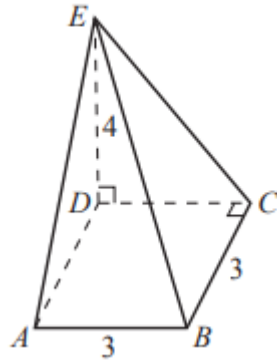


The diagrams show a solid pyramid L cut into two parts, M and N, by a plane parallel to its base. The base of pyramid L is a rectangle 9cm by 12cm. The perpendicular height of pyramid L is 30cm.

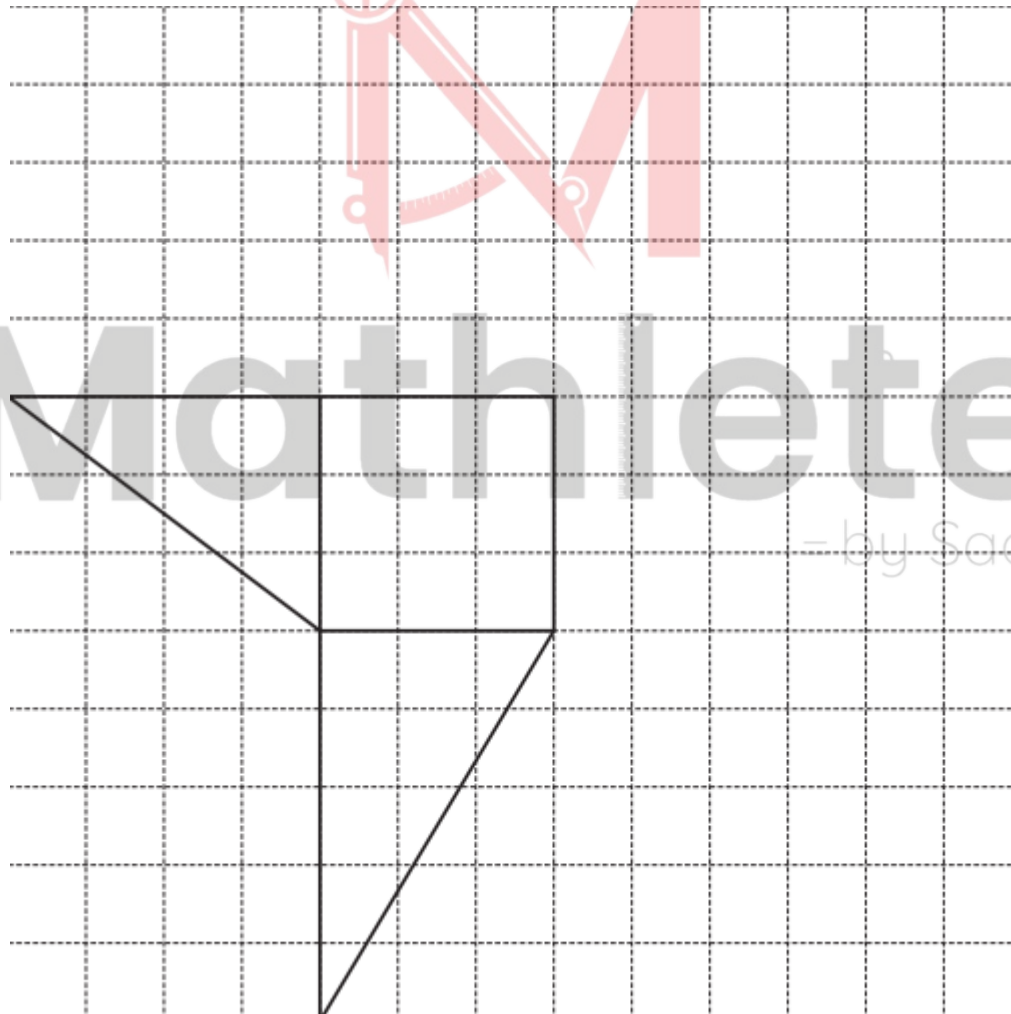
- a. Work out the volume of pyramid L. [1]
- b. The perpendicular height of pyramid M is $\frac{1}{3}$ of the perpendicular height of pyramid L.
- Express the volume of M as a fraction of the volume of L.
 - Calculate the volume of the solid N.

9. S18/qp12/q17

The diagram shows a pyramid. The square base, ABCD, has an edge of 3cm. The base is horizontal, and vertex E is vertically above D, where $ED = 4$ cm.



On the grid below, complete the accurate drawing of a net of the pyramid. Do not draw outside the grid.



10. S21/qp42/q8(b)(i) (0580)

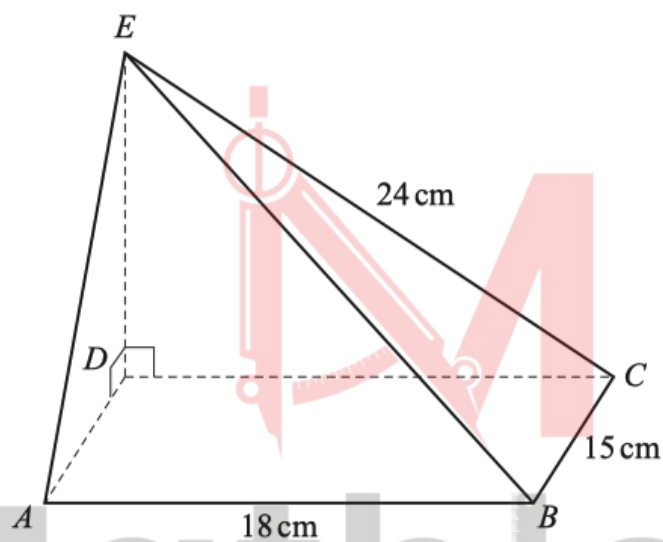
The diagram shows a solid pyramid with a rectangular base ABCD. E is vertically above D. Angle EDC = angle EDA = 90°. AB = 18cm, BC = 15cm and EC = 24 cm.

The pyramid is made of wood and has a mass of 800g. Calculate the density of the wood. Give the units of your answer.

[The volume, V , of a pyramid is $V = \frac{1}{3} \times \text{area of base} \times \text{height}$]

[Density = mass $\div$ volume]

[5]



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Answer Key

1. S10/qp21/q12(b)(ii)

(ii) (a) $VZ^2 = 15^2 - 6^2$ $VZ = 13.7$ to 13.75 cm	M1 A1	Value of 6 and correct use of Pythagoras
(b) 766 cm^3 (Accept $762 - 766$)	B1 ft [12]	ft $\frac{1}{3} \times$ their (b)(i)(a) $\times$ their (b)(ii)(a)

2. W10/qp22/q9(a)(b)

9	(a) 5 (b) (i) 20.8, $20\frac{5}{6}$ (ii) 21.6 (iii) 2.89 (cm) ft	1	2 B1 for $\frac{1}{2} \times 5 \times 5$ 3 M2 for $\frac{1}{2}$ their $(\sqrt{5^2 + 5^2})^2 \sin 60$ or B1 for $x^2 = 5^2 + 5^2$ oe or M1 for $\frac{1}{2} \times$ their $x^2 \times \sin 60$ 3ft ft for $(3 \times \text{their (i)}) \div \text{their (ii)}$ evaluated or M2 for $h = \frac{3 \times \text{their (i)}}{\text{their (ii)}}$ or M1 for $\frac{1}{2} \times \text{their (ii)} \times h = \text{their (i)}$
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3. S11/qp23/q21 0580

21	96 www	5	M1 $3^2 + 4^2$ A1 5 M1 $\frac{1}{2} \times 6 \times "5"$ (= 15) M1 $4 \times$ their triangle area $+ 6^2$
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4. S11/qp42/q7(a) 0580

7 (a)	87.5 (87.45 to 87.52) www 4	4	M1 for $\frac{1}{2} \times 2.5 \times 9.5$ soi by 11.875 or 71.25 and M2 for $\frac{1}{2} \times 2.5^2 \times \sin 60 \times 6$ oe (16.23 to 16.24) or M1 for $\frac{1}{2} \times 2.5^2 \times \sin 60$ (2.706..) = by Saad or 1 trapezium (8.1189..)
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5. W11/qp22/q7(b)

(b) (i) 111	1ft	Accept $4 \times$ their (a)(i) ft
(ii) 60.3	3ft	M1 for $(VF^2 =) 8^2 - (\text{their (a)(iii)})^2$ A1 for $(VF =) 6.53$ or ft soi SC1 for $\frac{1}{3} \times \text{their (a)(i)} \times \text{their } VF$

6. S15/qp23/q18

18	912 or 912.2...	5	M4 for $4 \times 0.5 \times 20 \times \sqrt{8^2 + 10^2} + 20 \times 20$ or better or M3 for $4 \times 0.5 \times 20 \times \sqrt{8^2 + 10^2}$ or better or M1 for $\sqrt{8^2 + 10^2}$ and M1 for $0.5 \times 20 \times \sqrt{8^2 + 10^2}$ and M1 for 20×20
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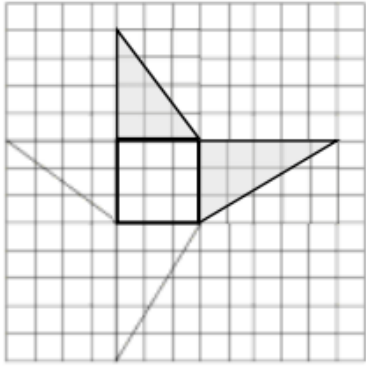
7. W16/qp21/q6

6 (a)	$[DT =]10.8 \text{ or } 10.816 \text{ to } 10.82$	2	M1 for $DT^2 = 6^2 + 9^2$ oe
(b)	139 or 139.2 to 139.3	3	B1 for $BT = 10$ M1 for sum of areas of four triangles seen, with at least 3 of the following correct: $\frac{1}{2} \times 8 \times 6$, $\frac{1}{2} \times 9 \times 6$, $\frac{1}{2} \times 8 \times \text{their } DT$, $\frac{1}{2} \times 9 \times \text{their } BT$
(c)	504	2	M1 for $9 \times 8 \times 5$ or $\frac{1}{3} \times 9 \times 8 \times 6$
(d)	50.7° final answer	3	M1 for finding an acute angle in triangle <i>THG</i>. e.g. $\tan [...] = \frac{11}{9}$ or $\tan [...] = \frac{9}{11}$ A1 for 50.7[...]° or 39.28 to 39.3°

8. S17/qp11/q15(a)

15(a)	1080	1	
15(b)(i)	$\frac{1}{27}$ oe	1	
15(b)(ii)	1040	2	M1 for <i>their</i> $1080 \times \text{their } \frac{1}{27}$ or 40
15(a)	1080	1	

9. S18/qp12/q17

17(a)	Correct net 	2	B1 for one correct triangle in correct position
17(b)	36 nfw	2	M1 for area of triangle $= \frac{1}{2} \times 3 \times 4$ or $\frac{1}{2} \times 3 \times 5$ soi

10. S21/qp42/q8(b)(i) 0580

8(b)(i)	0.559 to 0.56[0...]	B4	M2 for $\frac{1}{3} \times 18 \times 15 \times \sqrt{24^2 - 18^2}$ isw conversion or M1 for $h^2 + 18^2 = 24^2$ oe or better M1 for figs 800 ÷ figs <i>their</i> volume isw
	g/cm^3 or g cm^{-3} final answer	B1	

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